

Chapter 1

Notation

The study of nonlinear continuum mechanics will require a large number of concepts and mathematical notation to express those concepts. As a point of departure for our discussion, we highlight the main variables and their brief description, grouped by order of presentation.

The reader approaching the topic for the first time will want to scan this section, without concern as to the meaning of the variables, since this will unfold in subsequent sections. The experienced reader will gain an quick appreciation for the notation conventions used in this manuscript and use this section as a reference.

Tensors

a	zeroth order (scalar)
$\mathbf{A}$	first order (vector) in the reference configuration
A_I	first order (vector) components in the reference configuration
$\mathbf{a}$	first order (vector) in the current configuration
a_i	first order (vector) components in the current configuration
$\mathbf{A}$	second order (matrix) in the reference configuration
$\mathbf{a}$	second order (matrix) in the current configuration
A_{IJ}	second order (matrix) components in the reference configuration
a_{ij}	second order (matrix) components in the current configuration
$\mathbb{A}$	fourth order in the reference configuration

$\mathbb{A}_{IJKL}$	fourth order components in the reference configuration
$\mathfrak{a}$	fourth order in the current configuration
$\mathfrak{a}_{ijkl}$	fourth order components in the current configuration
$\hat{\mathbf{E}}_I$	unit basis vector in the reference (material) configuration
$\hat{\mathbf{e}}_i$	unit basis vector in the current (spatial) configuration
$\hat{\mathbf{b}}_i$	unit basis vector in the body B reference frame
o	origin in spatial configuration
O	origin of body B and location of the body reference frame $\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \hat{\mathbf{b}}_3$

Motion

$\mathbf{X}$	position of material particle in reference configuration
$\mathbf{x}$	position of material particle in current configuration
$\mathbf{U}$	displacement vector from $\mathbf{X}$ to $\mathbf{x}$, objective
φ	motion of material particle $\mathbf{X}$
$\varphi_*[\cdot]$	push forward operator
$\varphi_*^{-1}[\cdot]$	pull back operator
$\mathbf{v}$	velocity of material particle in current configuration, non-objective
$\mathbf{a}$	acceleration of material particle in current configuration, non-objective
$\nabla_0(\cdot)$	gradient operator in the reference configuration
$\nabla(\cdot)$	gradient operator in the current configuration
$\nabla_0 \mathbf{U}$	displacement gradient in the reference configuration, $\nabla_0 \mathbf{U} = \mathbf{F} - \mathbf{1}$
$\nabla \mathbf{u}$	displacement gradient in the current configuration, $\nabla \mathbf{u} = \mathbf{1} - \mathbf{F}^{-1}$
$\mathbf{F} = \nabla_0 \varphi$	deformation gradient, $\mathbf{F} = \mathbf{VR} = \mathbf{RU}$, non-symmetric, two-point tensor, objective (transforms as a vector, not a tensor)
$\mathbf{R}$	rotation, objective (transforms as a vector, not a tensor)
$\mathbf{V}$	left-stretch, objective

U right-stretch, objective

Strain

C right Cauchy-Green, $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, symmetric, positive definite, objective

b left Cauchy-Green, $\mathbf{b} = \mathbf{F} \mathbf{F}^T$, symmetric, positive definite, objective

E Green-Lagrange, $\mathbf{E} = \frac{1}{2} (\mathbf{C} - \mathbf{1})$, objective

e Euler-Almansi, objective

$\mathbf{E}_{(m)}$ Seth-Hill

λ_I principal stretch ratio

ϵ infinitesimal

$\dot{\mathbf{F}}$ material velocity gradient

$\nabla \mathbf{v}$ spatial velocity gradient, non-symmetric, non-objective, covariant

d stretch rate (spatial rate of deformation, rate of stretch), symmetric, objective, covariant

w rotation rate (spin, rate of rotation, vorticity), skew-symmetric, non-objective, covariant

Stress

$\boldsymbol{\sigma}$ Cauchy, objective

$\hat{\boldsymbol{\sigma}}$ Corotational Cauchy stress

$\dot{\boldsymbol{\sigma}}$ Cauchy stress rate, non-objective

$\overset{\nabla}{\boldsymbol{\sigma}}$ Jaumann-Zaremba stress rate, objective

$\square \boldsymbol{\sigma}$ Green-McInnis-Naghdi stress rate, objective

$\overset{\circ}{\boldsymbol{\sigma}}$ Truesdell stress rate, objective

$\boldsymbol{\tau}$ Kirchhoff

B Biot (Jaumann)

P Piola-Kirchhoff One, objective (transforms as a vector, not a tensor)

Σ Piola-Kirchhoff Two, objective

$\mathbf{A}$ Almansi

Energy

P pressure

V volume

$\mathcal{V}$ specific volume

T temperature

S entropy

ρ_0 density in the reference configuration

ρ density in the current configuration

m mass

E total energy

$\mathcal{E}$ total specific energy

U internal energy

K kinetic energy

G potential energy for external forces

$\mathcal{F}$ Helmholtz free energy

$\mathcal{G}$ Gibbs free energy

$\mathcal{H}$ enthalpy

Fluids

$\bar{\rho}_0$ constant density in the reference configuration

Elasticity

E elastic modulus

ν Poisson ratio

B bulk modulus

G shear modulus

Plasticity $\varepsilon^{\mathbf{e}}$ elastic strain $\varepsilon^{\mathbf{p}}$ plastic strain $\dot{\varepsilon}$ strain rate $\dot{\varepsilon}^{\mathbf{p}}$ plastic strain rate Φ yield function $\dot{\gamma}$ plastic multiplier H hardening modulus**Contact** $\mathbf{a}$ a $\mathbf{b}$ b**Shock** $\mathbf{a}$ a $\mathbf{b}$ b

4.7 Simple Shear

Figure 4.7 illustrates **simple shear**, with relative motion of planes normal to the vertical axis. For all configurations (reference and current):

- Horizontal fibers remain horizontal.
- The length of horizontal fibers remains constant.
- The vertical distance between the horizontal fibers remains constant.
- The body's volume is constant. The deformation is isochoric since $\det(\mathbf{F}) = 1$, see (4.71).

The relative motion is characterized by the non-dimensional ratio of length a per unit height, where $a \in \mathbb{R} \subset [0, \infty)$. The angle, $\gamma \in \mathbb{R} \subset [0, \pi/2)$, subtended by simple shear is $\gamma = \arctan(a)$. In the limit as $a \mapsto \infty$, the shear angle $\gamma \mapsto \pi/2$. For small values of a , the small-angle assumption is valid, with $\tan(\gamma) \approx \gamma \approx a$.

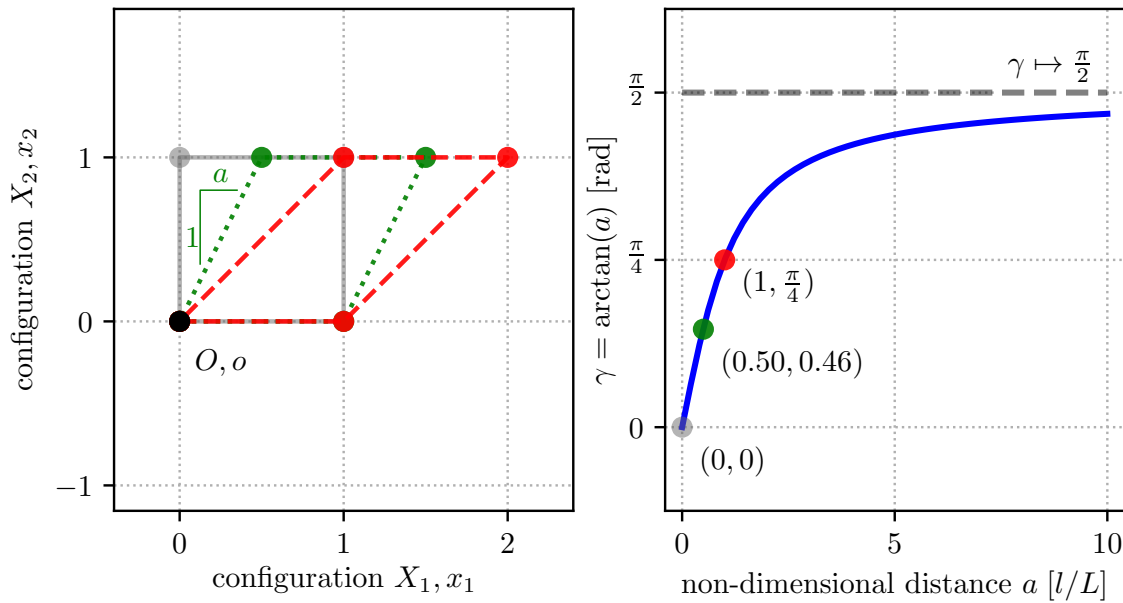


Figure 4.7: (Left) **Simple shear** of a unit cube in the reference configuration (gray) and two current configurations (dotted, green and dashed, red) and (right) with the shear angle, γ , created for all configurations that are parameterized by $a \in [0, 10]$.

In Fig. 4.7 (left), the horizontal displacement per unit height, a , is illustrated by the dotted, green configuration. The (right) figure also shows the shear angle $\gamma =$

$\arctan(a)$, with the three (a, γ) pairs plotted in a color that matches the corresponding deformation in the left figure.

4.7.1 Rate-Independent

The simple shear deformation φ in rate-independent form is

$$\varphi(\mathbf{X}) = (X_1 + aX_2) \mathbf{e}_1 + X_2 \mathbf{e}_2 + X_3 \mathbf{e}_3. \quad (4.69)$$

The deformation gradient $\mathbf{F}$ is

$$\mathbf{F} = \mathbf{e}_i \frac{\partial \varphi_i}{\partial X_J} \mathbf{E}_J = \mathbf{e}_1 \mathbf{E}_1 + a \mathbf{e}_1 \mathbf{E}_2 + \mathbf{e}_2 \mathbf{E}_2 + \mathbf{e}_3 \mathbf{E}_3, \quad (4.70)$$

$$= \{\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3\} \begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left\{ \begin{matrix} \mathbf{E}_1 \\ \mathbf{E}_2 \\ \mathbf{E}_3 \end{matrix} \right\} \iff F_{iJ} = \begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (4.71)$$

Note that the volume remains constant for all deformations since $\det(\mathbf{F}) = 1$. The deformation thus belongs to the group of isochoric motions (see Fig. 4.1). The right Cauchy-Green strain tensor, $\mathbf{C} = \mathbf{F}^\top \mathbf{F}$, is

$$\mathbf{C} = \mathbf{E}_1 \mathbf{E}_1 + a (\mathbf{E}_1 \mathbf{E}_2 + \mathbf{E}_2 \mathbf{E}_1) + (a^2 + 1) \mathbf{E}_2 \mathbf{E}_2 + \mathbf{E}_3 \mathbf{E}_3, \quad (4.72)$$

$$\iff C_{IJ} = \begin{bmatrix} 1 & a & 0 \\ a & a^2 + 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (4.73)$$

The Green-Lagrange strain tensor is simply $\mathbf{E} = \frac{1}{2} \mathbf{C}$. The left Cauchy-Green strain tensor, $\mathbf{b} = \mathbf{F} \mathbf{F}^\top$, is

4.7.2 Rate-Dependent

The simple shear deformation φ in rate-dependent form is

$$\varphi(\mathbf{X}, t) = (X_1 + a(t)X_2) \mathbf{e}_1 + X_2 \mathbf{e}_2 + X_3 \mathbf{e}_3. \quad (4.74)$$

The deformation gradient $\mathbf{F}$ is unchanged from (4.71). Additionally, the inverse and rate, respectively, are found to be

$$\mathbf{F}_{iJ}^{-1} = \begin{bmatrix} 1 & -a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \dot{\mathbf{F}}_{iJ} = \begin{bmatrix} 0 & \dot{a} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (4.75)$$

With $\mathbf{C}$ unchanged from (4.73), and $\dot{\mathbf{C}} = \dot{\mathbf{F}}^\top \mathbf{F} + \mathbf{F}^\top \dot{\mathbf{F}}$,

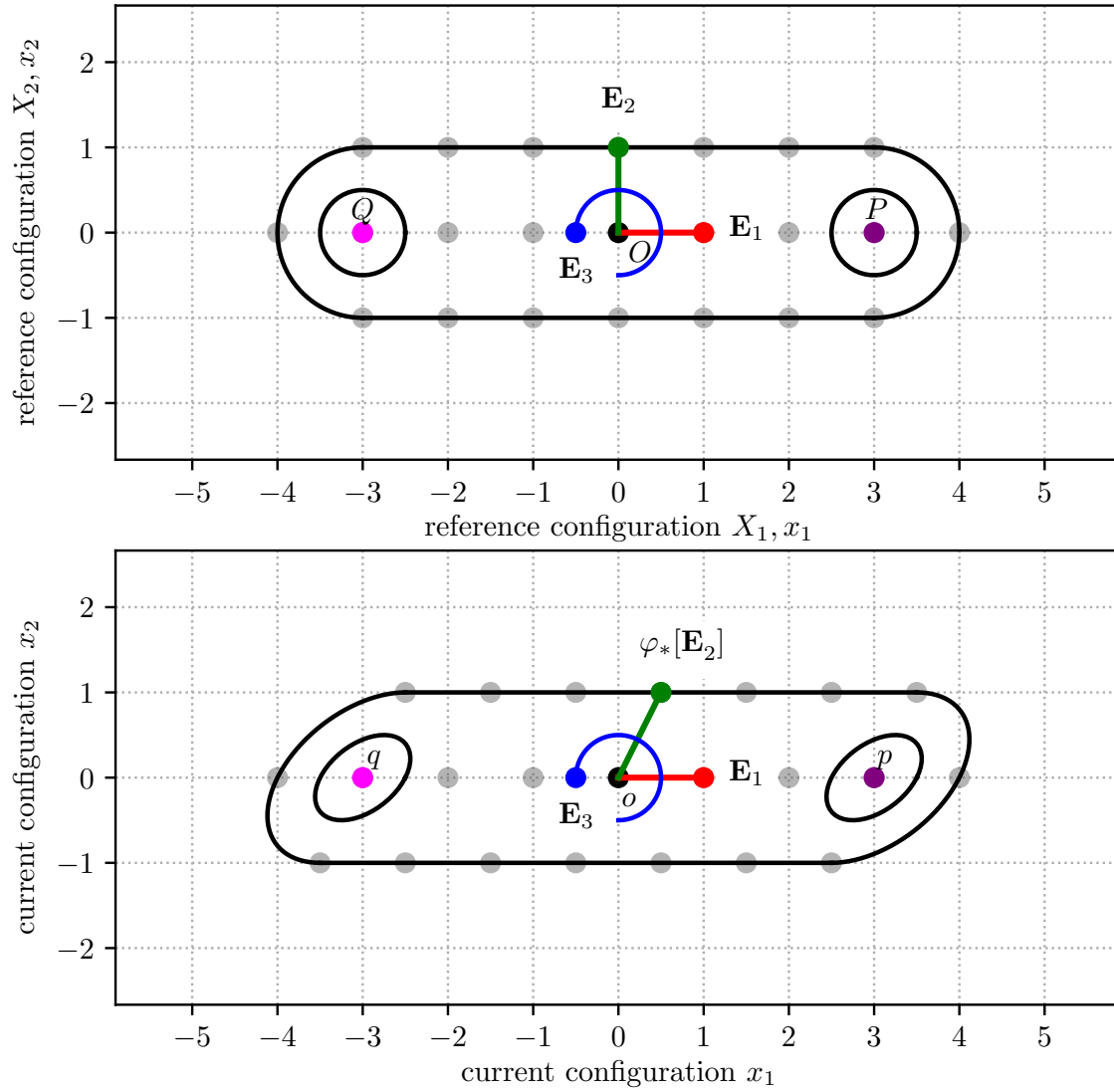


Figure 4.8: Simple shear of body from the reference configuration B to the current configuration $\varphi(B)$.